

ORGANIC MATTER DEGRADATION DURING WEATHERING OF PALEOZOIC BLACK SHALES – A TG-MS STUDY

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Mineralization of dispersed organic matter (kerogen) from ancient black shales during weathering is one of the major factors controlling the atmospheric O₂-level (Berner 2001). It has been shown that weathering causes significant changes in amount and composition of the residual organic matter in such rocks (e.g. Littke et al. 1991) and that biological processes play an important role in the transformation of ancient organic matter (Petsch et al. 2001). Man-made exposure of rocks from reducing conditions of the subsurface to atmospheric conditions by mining forces these processes and the effects of that additional oxidation are not yet established.

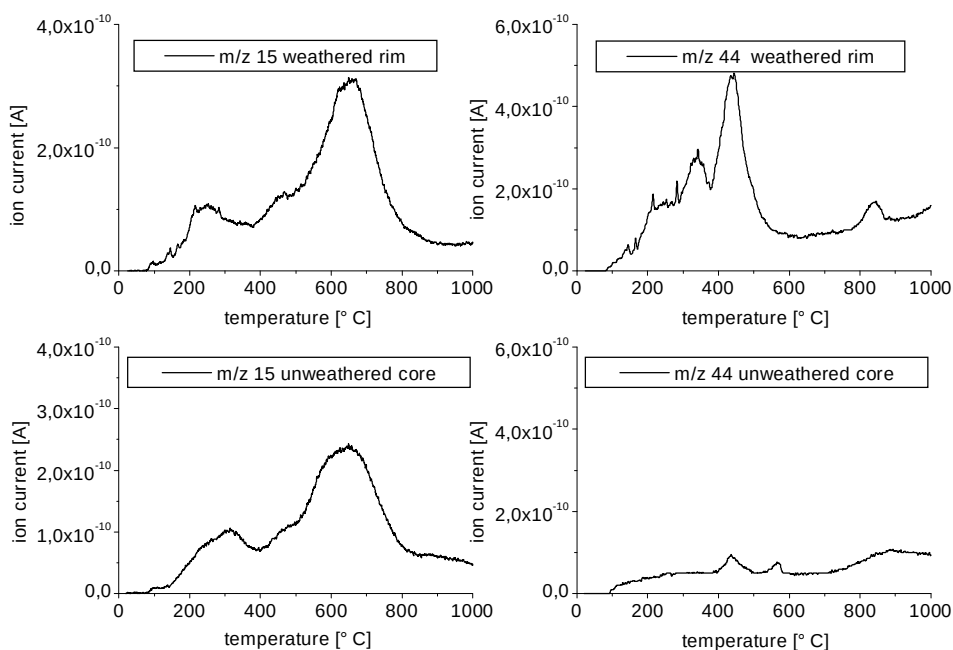


Fig. 1: Different hydrocarbon (m/z 15) and carbon dioxide (m/z 44) degassing due to weathering

We present TG-MS (DEGAS, Schmidt and Heide 2001) data from oxidative weathered Lower Silurian aluminous shales of the Berga and Schwarzburg Anticline, East Thuringia, Germany. These studies are based on the temperature resolved analysis of low molecular weight fragments during vacuum pyrolysis (heating rate 10 K/min) and the simultaneous detection of inorganic volatiles (e.g. H₂O, CO₂, CO, N₂, and SO₂). Especially the evolution of CO₂ from carboxylic acids reveals valuable information about oxidation processes on the residual kerogen fraction in weathered rock specimens (Fig. 1).

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